

General Directions: Complete all of the following questions. A perfect examination will earn 105 out of 100 points. Read the directions for each section carefully. Failure to comply with the directions may result in a loss of points. For example, writing two letters in one blank will receive 0.

Due Date: 11:59 p.m., Wednesday, October 30, 2024 on Canvas

- This is an open-book, open-notes, untimed examination.
- Do not discuss this exam with anyone else (including bots), except for the instructor.
- If you are uncertain what a question is asking, you may contact the instructor (butlera@iastate.edu) for clarification; or you may write a brief explanation describing how you interpret the question.
- Save your completed exam in a standard document (.docx, .pdf, .rtf) or picture format. Or you may type your answers using Canvas's text submission box.
- Submit your exam on Canvas by the due date. Consult the syllabus for the late policy.

Note: The name(s) in parentheses refers to the reading(s) associated with the question.

Section 1. Multiple Choice. Write the letter of the one best option in each blank. NOTE: The same letter may be used more than once; not every letter must be used. (3 points each; 93 points total)

- _____ 1. **True or False:** According to boyd and Crawford: If a claim involves numerical data, the claim must be "objective." (boyd and Crawford)
- A. True: numbers are always objective
 - B. False
- _____ 2. Some studies show that when students use AI to help them answer quizzes, the students overestimate their own knowledge (independent of the quiz).
Question: According to Messeri and Crockett, what type of illusion is this an example of? (Messeri and Crockett)
- A. Illusion of explanatory depth
 - B. Illusion of exploratory breadth
 - C. Illusion of objectivity
 - D. Illusion of anthropomorphism
- _____ 3. According to Mittelstadt: Which of the following explanations would address *transparency*? (Mittelstadt)
- A. A functional explanation of the algorithm's calculations
 - B. A behavioral explanation of the correlations between inputs and outputs
 - C. Disclosure of how the "black box" operates
 - D. Account of whose data were used in creation of the algorithm

- _____ 4. What word or phrase does Green use to refer to what Broussard calls “technochauvinism”? (Broussard and Green)
- A. Human oversight
 - B. Automation bias
 - C. Flexibility
 - D. Red line
- _____ 5. Consider the following argument: Hiring algorithms may have biases. But so do humans. And humans biases are even worse! So we should choose the less biased hiring method and use hiring algorithms instead of humans. **Question:** How does Ajunwa reply to this argument? (Ajunwa)
- A. Hiring software is too expensive
 - B. Hiring algorithms have human biases built into them
 - C. Applicants feel more comfortable with a human interviewer, rather than a faceless algorithm
 - D. None of the above: Ajunwa agrees with this argument
- _____ 6. **True or False:** According to Green, municipal governments should never have humans and algorithms work together. (Green)
- A. True: human judgments will always introduce mistakes, so decision making should always be left to algorithms
 - B. True: it is impossible for humans and algorithms to work together, because humans are flexible and algorithms are inflexible
 - C. False: for high-stakes decisions with demonstrably reliable algorithms, it may be appropriate for algorithms and humans to work together
 - D. False: human monitors can be trained to overcome undue deference to an algorithm’s outcomes
- _____ 7. Google Search collects users’ search terms and analyzes patterns and search behavior. **Question:** Why does Zuboff not describe users as Google’s *workers*? (Zuboff)
- A. Google Search’s users do not produce anything useful
 - B. Google Search’s users do not pay to use Google’s services
 - C. Google Search’s users pay to use Google’s services
 - D. Google does not pay users to use Google Search
- _____ 8. According to Zuboff, what distinguishes “surveillance capitalism” from other types of capitalism? (Zuboff)
- A. Surveillance capitalism aims towards high productivity
 - B. Surveillance capitalism is not concerned with expanding services
 - C. Surveillance capitalism aims to maximize profit
 - D. Surveillance capitalism’s profits do not come from producing more gadgets faster

- _____ 9. **True or False:** According to Zuboff, Google’s User Profile Information (UPI) includes only those behavioral data that the users consent to provide. (Zuboff)
- A. True: UPIs are limited to information the user supplies voluntarily
 - B. False
- _____ 10. **True or False:** According to Zuboff, why is “digital exhaust” a euphemism? (Zuboff)
- A. It implies that users are too tired to care about the data, when in fact users care a lot about the data
 - B. It implies that the user discards the data, when the data are left behind intentionally by the user
 - C. It implies that nothing of any value is collected, when in fact the data are very valuable
 - D. It implies that the data reveal everything that there is to be known about the user, when in fact the data are limited to what the user happened to search
- _____ 11. **True or False:** According to Benn and Lazar, establishing that a practice causes harm on an *interactional approach* requires showing that at least one individual is harmed. (Benn and Lazar)
- A. True: interactional approaches need to show harms to at least one individual
 - B. False
- _____ 12. **True or False:** According to Benn and Lazar’s structural approach, if there is someone who does not modify their behavior because of targeted messaging, then that unaffected person proves that the targeted messaging is not manipulative. (Benn and Lazar)
- A. True: a single person’s behavior can disprove a structural claim
 - B. False
- _____ 13. When Benn and Lazar claim that the manipulation by automated influence preys on our propensity to Tragedy of the Commons, what is their point? (Benn and Lazar)
- A. Automated influence leads users to choose options that are worse for them than they would have chosen had they not been manipulated by automated influence
 - B. Automated influence distracts us from we most care about
 - C. Automated influence capitalizes on our stronger drive to promote our own interest over the weaker drive to cooperate with others, even in situations where cooperation would produce overall better outcomes
 - D. Automated influence always makes everybody worse off
- _____ 14. What kind of liberty would Saetra classify the First Amendment’s guarantee that the government will not restrict a person’s speech? (Saetra)
- A. Positive liberty
 - B. Negative liberty

NAME: _____

- _____ 15. Which of the following reasons would Saetra offer for why nudges are bad? (Saetra)
- A. Nudges intervene in decision-making without the person's having first asked for assistance
 - B. Nudges reduce the number of options available to a person to choose from
 - C. Nudges can be used for morally bad goals too
 - D. Nudges help people to learn what is in their best interest
- _____ 16. Which of the following would Saetra say is least likely to be coercive? (Saetra)
- A. A schoolyard bully tackles a student to take the student's lunch money
 - B. Amazon repeatedly sends notifications about the drop in price on an item you once browsed
 - C. A Bey (member of the Bey Hive, the name for Beyonce's fans) offers reasons for why Beyonce's new country album is genuinely country music
 - D. Your significant other threatens to withhold affection unless you comply with their request
- _____ 17. On what point does Zuboff agree with Benn and Lazar? (Zuboff and Benn and Lazar)
- A. Each individual has a right to unilateral control over their data
 - B. Automated influence is always morally wrong
 - C. Users consent to sharing their data when they use a service
 - D. Users have less resources, knowledge, and power than companies that use automated influence, because users do not control the means of prediction
- _____ 18. **True or False:** Benn and Lazar would say that anyone who signs an end user license agreement (EULA) has consented to collection of their personal data and cannot be exploited. (Benn and Lazar)
- A. True: once a EULA has been signed, there is no possible way for data collection to be exploitative
 - B. False
- _____ 19. **True or False:** According to Benn and Lazar, because users' data taken together generate something very valuable, a fairer (more just) system would allow users to participate in deciding how their data are used.
- A. True: those who contribute to the surplus of resources ought to have a voice in how those resources are used
 - B. False
- _____ 20. **True or False:** According to Moriarty, online price discrimination is always unfair. (Moriarty)
- A. True: there are no circumstances when online price discrimination is fair
 - B. False

NAME: _____

- _____ 21. Suppose that a store offers a 10% discount to U.S. Veterans. **Question:** What degree of price discrimination is that? (Moriarty)
- A. First-degree price discrimination
 - B. Second-degree price discrimination
 - C. Third-degree price discrimination
 - D. Trick question: this is not price discrimination
- _____ 22. Why does Moriarty think price discrimination generally is not exploitative? (Moriarty)
- A. Sellers do not always use personal data about the buyer in online transactions
 - B. Conducting business online does not create vulnerability for the seller
 - C. Typically the buyer sets their reservation price
 - D. Sellers and buyers have the same power, knowledge, and resources
- _____ 23. According to Moriarty, if a seller has access to personal information about the buyer, is that automatically an unfair advantage? (Moriarty)
- A. Yes, always, even if that knowledge is disclosed
 - B. Yes, but only if the seller obtains informed consent from the buyer first
 - C. No, because if that advantage is disclosed, the buyer can decide not to participate in the transaction
 - D. No, because the information may be inaccurate
- _____ 24. Suppose that a store discounts the cost per item when more than three of those items are purchased. **Question:** Would Moriarty regard this as unfair? (Moriarty)
- A. Yes, fairness requires that the same price be offered to everybody
 - B. Yes, encouraging people to buy in bulk depletes the supply for other buyers
 - C. Yes, not everyone can afford to buy in bulk
 - D. No, as long as the discount is made clear to all buyers
- _____ 25. What kind of harm would Solove classify the following as: the government collects your DNA as a child in case you are kidnapped, and later in life arrests you for a crime on the basis of DNA evidence. (Solove)
- A. Secondary Use
 - B. Chilly Climate
 - C. Aggregation
 - D. Exclusion
- _____ 26. What kind of harm would Solove classify the following as: a credit rating agency assigns a person a creditworthiness score based on the agency's secret proprietary algorithm. (Solove)
- A. Secondary Use
 - B. Chilly Climate
 - C. Aggregation
 - D. Exclusion

- _____ 27. **True or False:** According to Solove, our interest in privacy is limited to concealing bad things about ourselves.
- A. True: the only reason to care about privacy is to prevent discovery of discreditable information
 - B. False
- _____ 28. Does Solove rely on the existence of chilling effects to conclude that widespread government surveillance is wrong?
- A. Yes, because surveillance reduces the range of viewpoints people are willing to express
 - B. Yes, because surveillance prevents people from joining protests against the government
 - C. No, it is difficult to prove that surveillance causes people not to exercise their First Amendment rights
 - D. No, there are no chilling effects from government surveillance
- _____ 29. **True or False:** According to Solove, government wiretapping of a citizen's phone is always morally wrong and can never be justified. (Solove)
- A. True: there are no circumstances in which wiretapping can be justified
 - B. False
- _____ 30. I visit a website to download a printer driver. In the middle of the page is a big neon green button labeled "Download" associated with sponsored content, not the printer driver. The button to download the driver is much lower on the page, in a much smaller font. **Question:** How would Richards and Hartzog classify this pathology of consent? (Richards and Hartzog)
- A. Unwitting consent (not knowing)
 - B. Coerced consent (not voluntary)
 - C. Incapacitated consent
- _____ 31. **True or False:** According to Richards and Hartzog: To avoid the pathologies of consent, the only change that is required for any End User License Agreement (notice) is in its wording. (Richards and Hartzog)
- A. True: the wording of notices is the only source of pathologies of consent
 - B. False: other circumstances of notice or consent may produce pathologies

NAME: _____

Section 2. Matching. In each blank, write the letter of the **one** best option (A or B). NOTE: The same option may be used more than once. Not every option must be used.
(3 point per blank; 12 points total)

What type of normative ethical theory does the described judgment or argument employ?

- A. Consequentialist (including utilitarianism)
- B. Deontological

- _____ 32. According to Zuboff: Google's User Profile Information strategy is morally wrong, because it treats users not as ends-in-themselves, but as mere means for selling ads (Zuboff)
- _____ 33. As reported by Saetra: Sunstein and Thaler justify nudges because they produce the greatest overall welfare (Saetra)
- _____ 34. According discussed by Benn and Lazar: Google's data collection is manipulative, because it undermines your ability to decide for yourself (Benn and Lazar)
- _____ 35. According to Solove: the Nothing to Hide Argument, which weighs the harm to the individual's privacy against the benefit of society's security (Solove)